UNIT-4

1. Find addition of polynomials $3x^2+2x+4$ and $2x^2+3x+2$.
2. Find addition of polynomial $4y^2+3y+1$ and $-2y^2-y+4$.
3. Find addition of polynomial $5x^5+6x^3+8x+100$ and $10x^2+11x+11$.
4. Find addition of polynomial x^2+x+1 and x^3-x-2.
5. Subtract $4x^3+3x^2+2x+3$ from $5x^3+4x^2+7$.
6. Subtract $-3x^2+50x-2$ from $4x^3-x^2+7$.
7. Subtract $x-5$ from $4x^4-5x^3+6x^2+8$.
8. Subtract x from x^2+1.
9. Subtract x^2-2 from x^4+x^2+1.
10. Use the Urdhva Triyabhyam sutra to multiply $2x+1$ and $3x+3$.
11. Use the Urdhva Triyabhyam sutra to multiply $4x+2$ and $2x+5$.
12. Use the Urdhva Triyabhyam sutra to multiply $2x+1$ and $3x+3$.
13. Use the Urdhva Triyabhyam sutra to multiply $4x^2+2x+3$ and $3x^2+3x+2$.
14. Use the Urdhva Triyabhyam sutra to multiply $7x^2+8x+2$ and $8x^2+6x+5$.
15. Use the Urdhva Triyabhyam sutra to multiply $2x^2+3x+4$ and $5x^2+6x+7$.
16. Use the Urdhva Triyabhyam sutra to multiply x^2+5 and $x^2-11x+6$.
17. Use the Urdhva Triyabhyam sutra to multiply $2x-5$ and $x-1$.
18. Use the Urdhva Triyabhyam sutra to find square of $2x-1$.
19. Use the Urdhva Triyabhyam sutra to find square of $5x^2-2x-1$.
20. Use the Urdhva Triyabhyam sutra to multiply $(3x-2),(x-1)$ and $(3x+5)$.
21. Use the Urdhva Triyabhyam sutra to find cube of $2x+1$.
22. Use the Urdhva Triyabhyam sutra to find cube of $x-2$.
23. Use the Urdhva Triyabhyam sutra to multiply $(2x+1)$,$(3x+3)$ and $(x-5)$
24. Use the concept of Paravartya to find $\frac{2x^2+4x+7}{x+2}$.
25. Use the concept of Paravartya to find remainder and quotient of $\frac{3x^3+2x^2+5x+9}{x-1}$.
26. Use the concept of Paravartya to find remainder and quotient of $\frac{4x^2-4}{x-1}$.
27. Use the concept of Paravartya to find remainder and quotient of $\frac{x^2+1}{x-1}$.
28. Use the concept of Paravartya to find remainder and quotient of $\frac{x^5+x^3+x-1}{x-5}$.
29. Use the concept of Paravartya to find remainder and quotient of $\frac{x^3-1}{x+1}$.

30. Use the concept of Paravartya to find remainder and quotient of $\frac{x^2-2x+1}{x-1}$.

31. Factorize quartile $4x^2 + 12x + 5$ use the concept of Anurupyena and Adyamayenantyamanteya.

32. Factorize quartile $15x^2 - 14xy - 8y^2$ use the concept of Anurupyena and Adyamayenantyamanteya.

33. Factorize quartile $3x^2 + 14x - 5$ use the concept of Anurupyena and Adyamayenantyamanteya.

34. Factorize quartile $6x^2 - 23x + 7$ use the concept of Anurupyena and Adyamayenantyamanteya.

35. Factorize quartile $8x^2 - 22x + 5$ use the concept of Anurupyena and Adyamayenantyamanteya.

36. Factorize quartile $12x^2 - 23xy + 10y^2$ use the concept of Anurupyena and Adyamayenantyamanteya.

37. Factorize polynomial $3x^2 + 7xy + 2y^2 + 11xz + 7yz + 6z^2$ use the concept of Lopana Sthapnabyam.

38. Factorize polynomial $12x^2 + 11xy + 2y^2 - 13xz - 7yz + 3z^2$ use the concept of Lopana Sthapnabyam.

39. Factorize polynomial $3x^2 + 6y^2 + 6z^2 + 11xy + 7yz + 6xz + 19x + 22y + 13z + 20$ use the concept of Lopana Sthapnabyam.

40. Factorize the polynomial $2x^3 - 3x^2 + 1$.

41.Factorize the polynomial $x^3 - 3x^2 - 9x - 5$.

UNIT-5

1. Find HCF(12,15,21) and LCM(12,15,21).
2. Find HCF(50,70,85) and LCM(50,70,85).
3. Find HCF(10,30,50,100) and LCM(10,30,50,100).
4. Find HCF(8,20,32) and LCM(8,20,32).
5. Find HCF(10,15,100) and LCM(10,15,100).
6. Find HCF(3,9,18,27) and LCM(3,9,18,27).
7. Find LCM(10,20,30) and HCF(10,20,30).
8. Find LCM(5,10,15) and HCF(5,10,15).
9. Find LCM(11,12,13) and HCF(11,12,13).
10. Find LCM(10,13,20) and HCF(10,13,20).
11. Find LCM(6,12,18) and HCF(6,12,18).
12. Solve following bivariant linear equations.

    1) $3x + 5y = 19$
    2) $2x + 3y = 12$

13. Solve following bivariant linear equations.

    1) $3x - 5y = 19$
    2) $-7x + 3y = -1$

14.Solve following bivariant linear equations.

1) $x + y = 8$
2) $2x - 3y = 1$

15.Solve following bivariant linear equations.

1) $11x + 15y = -23$
2) $7x - 2y = 20$

16.Solve following bivariant linear equations.

1) $2x + y = 5$
2) $3x - 4y = 2$

17.Solve following bivariant linear equations.

1) $5x - 3y = 11$
2) $6x - 5y = 9$

18.Solve following bivariant linear equations.

1) $11x + 6y = 28$
2) $7x - 4y = 10$

19.Solve following bivariant linear equations.

1) $6x + 7y = 8$
2) $19x + 14y = 16$

20.Solve following bivariant linear equations.

1) $12x + 8y = 7$
2) $16x + 16y = 14$

21.Solve following bivariant linear equations.

1) $12x + 78y = 12$
2) $16x + 96y = 16$

22.Solve equation $\frac{3x+4}{6x+7} = \frac{5x+6}{2x+3}$.

23.Solve equation $\frac{1}{x+7} + \frac{1}{x+9} = \frac{1}{x+6} + \frac{1}{x+10}$.

24.Solve equation $\frac{1}{x-7} + \frac{1}{x+9} = \frac{1}{x+11} + \frac{1}{x-9}$.

25.Solve equation $\frac{1}{x-8} + \frac{1}{x-9} = \frac{1}{x-5} + \frac{1}{x-12}$.

26.Solve equation $\frac{x-2}{x-3}+\frac{x-3}{x-4}=\frac{x-1}{x-2}+\frac{x-4}{x-5}$.

27.Solve equation $\frac{x}{x-2}+\frac{x-9}{x-7}=\frac{x+1}{x-1}+\frac{x-8}{x-6}$.

28.Solve equation $\frac{2x-3}{x-2}+\frac{3x-20}{x-7}=\frac{x-3}{x-4}+\frac{4x-19}{x-5}$.

29.Solve equation $\frac{3x-8}{x-3}+\frac{4x-35}{x-9}=\frac{2x-9}{x-5}+\frac{5x-34}{x-7}$.

30.Solve equation $\frac{3x-13}{x-4}+\frac{4x-41}{x-10}=\frac{2x-13}{x-6}+\frac{5x-41}{x-8}$.

31.Solve the equation $\frac{2}{2x+3}+\frac{3}{3x+2}=\frac{1}{x+1}+\frac{6}{6x+7}$.

32.Solve equation $\frac{3}{3x+1}-\frac{6}{6x+1}=\frac{3}{3x+2}-\frac{2}{2x+1}$.

33.Solve equation $\frac{3}{3x+1}+\frac{2}{2x-1}=\frac{3}{3x-2}+\frac{2}{2x+1}$.